

# Jason Firenze Trianto

Computer Science (Intelligent Systems), BINUS University

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## ABOUT ME

Computer Science student at BINUS University that specializes in Intelligent Systems and is focused on developing AI related projects. Experienced in developing both individual and team based projects around NLP/NER research, computer vision, classical machine learning, and web development. Currently seeking an AI Engineering internship to contribute to building and deploying AI systems.

**Key Achievements:** Authored and presented an IEEE format research paper comparing 4 models for allergen detection in product ingredient labels, accepted for publication at ICORIS 2026.

## EDUCATION

### BINUS University

2024-2028

Bachelor of Computer Science, Specializing in Intelligent Systems

**GPA: 3.5 / 4.00**

BINUS University, Jakarta

## PROJECTS

### AllergenScan - Indonesian Food Allergen Detection (NLP & OCR - Research)

**Purpose:** Help consumers and people with dietary restrictions or allergies catch hidden allergens on Indonesian food packaging where labels sometimes mix Indonesian and English and manual ingredient checking is slow and error prone.

**Result:** Built an end to end OCR + IndoBERT NER pipeline trained via distant supervision on a 3,500 sample dataset, achieved a micro F1 of 0.741 and a macro F1 of 0.720 across 8 allergen categories on 137 real product photos, outperforming dictionary, TF-IDF, and mBERT baselines on recall.

**Documentation:** [HuggingFace Link](#)

### WEAR - AI Wardrobe Organizer (Software Engineering - Team of 4)

**Purpose:** Help users manage their wardrobe and cut down daily decision fatigue by generating outfit recommendations based on occasion, season, and mood.

**Result:** Delivered frontend features such as dark mode, responsive UI, filters, and AI powered clothing image analysis for a full-stack app powered by Gemini's API key, contributed to a 68/68 passing Jest test suite, deployed live on Vercel with Supabase.

**Documentation:** [GitHub Link](#)

### Automated Contrast Correction System (Computer Vision - Team of 6)

**Purpose:** Automatically fix under and over exposed photos without manual parameter tuning, addressing the limitation of standard CLAHE requiring users to manually choose a clip limit and tile size.

**Result:** Built the Streamlit app integrating CLAHE with two trained ML models, a 96.3% accurate exposure classifier and a 99.6% accurate tile size predictor, the enhancement pipeline outperformed baseline Global Histogram Equalization by 2.0% in SSIM.

**Documentation:** [GitHub Link](#)

### Fruit & Vegetable Freshness Classifier (Machine Learning - Team of 4)

**Purpose:** Provide a fast, lightweight way to classify produce as fresh or rotten without relying on a heavier deep learning model.

**Result:** Engineered an 8,196 dimension HOG + HSV feature set reduced via PCA, tuned an SVM to 91% accuracy with under 16ms end-to-end inference latency, and added LIME-based explainability to visualize model decisions.

**Documentation:** [GitHub Link](#)

### **Crop Yield Predictor - AI Agritech System (API Integration - Team of 3)**

**Purpose:** Give farmers an accessible way to estimate crop yield and get practical guidance under unpredictable climate conditions, without requiring custom ML training or large historical datasets.

**Result:** Created an API integrated agritech app estimating yield per hectare from live weather, soil, and crop data, paired with a Gemini powered farming chatbot, led evaluation and revision of the deployed app and produced its presentation materials.

**Documentation:** [GitHub Link](#)

## **PUBLICATIONS**

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Rafifdiya, Ahmad Daffa Hidayatullah, **Jason Firenze Trianto**, Bakti Amirul Jabar, Rico Halim. "Distant Supervision for Indonesian Food Allergen NER: A Preliminary Evaluation with OCR Integration." Proceedings of ICORIS 2026.

## **TECHNICAL SKILLS AND OTHERS**

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**Languages:** Indonesian, English

**Certification and Training:** Azure AI Fundamental Completion, Google AI Professional Certificate

**Web and Design:** HTML, CSS, JavaScript, Figma, React, Next.js

**AI / Machine Learning:** Scikit-learn, OpenCV, Hugging Face Transformers, NLP / Named Entity Recognition, Feature Engineering, Model Evaluation, Prompt Engineering

**Tools and Platforms:** Git / GitHub, Streamlit, Vercel, Supabase, Google Colab / Jupyter